E-Commerce Fulfilment Analytics Take-Home Task

Time allowed: up to 4 hours

Background

You work for an online marketplace that ships physical goods across four UK regions (North, South, East, West). Three third-party couriers (Arrow, Bolt, Cyclone) handle deliveries. Refunds hurt profit, so management wants to understand where time and money leak out of the fulfilment pipeline.

You have been sent **three CSV files by email**:

- **orders.csv**: One row per order
- **shipments.csv**: One row per parcel shipped (an order may generate multiple parcels)
- **refunds.csv**: One row per order that was refunded (fully or partially)

Deliverables

Please submit all three of the following via the Ashby submission link you have received in the email

1. **Section 1 answers:**
    - A single text document (Word, PDF, or similar) with clear, numbered answers to Q1–Q8
    - *Include only text answers, no charts of visuals needed here*
2. **Section 2 answers:**
    - For each question, write your answer as text (bullet points or short paragraphs), and include the key charts or tables that helped you reach your conclusion
    - Combine text and charts as either a short slide deck, or Word/PDF document with embedded charts / tables
3. **Workings:**
    - A separate file (Python, SQL or Excel) covering both **Section 1 and Section2**
    - Please reference questions or slide numbers where applicable

Section 1

Provide clear, numbered answers (Q1–Q8) as single bullet points or short paragraphs. Only final answers are needed unless stated otherwise. Submit code working separately.

1. Which courier handled the most parcels?
2. What is the overall average shipping time (days) for delivered parcels?
3. Which region has the longest average shipping time?

4. Which product category generated the highest total order value?
5. What is the refund rate (number of orders refunded divided by all orders) for each product category? Which is worst?
6. Is the gap between best and worst category refund rates statistically significant? Briefly explain your approach and test
7. Which courier has the lowest percentage of their parcels that were returned AND refunded?
8. Calculate net revenue per parcel for each combination of courier and product category (eg. Arrow–Home, Cyclone–Clothing). Which combination yields the highest net revenue per parcel? State any assumptions

Section 2

For each question below:

- Write your answer as text (bullet points or a short paragraph).
- Include the key charts or tables that helped you reach your recommendation. These should clearly show the numbers or comparisons you used

Business questions to address:

1. **Single courier strategy**: if you had to select one courier for next quarter, which would you choose and why? Give at least two quantitative arguments, supported by visuals

2. **Segmented courier strategy**: how would you assign couriers to regions or product categories to improve performance? Justify your recommendation with data / visuals 3. **Operational improvements**: recommend two operational changes (beyond changing courier allocation) to reduce refund-related losses, using insights from your analysis

- Please also include a brief note of any key assumptions or data limitations you identified, and how they might affect your conclusions

Data Dictionary

Column	File(s)	Description
order_id	all	Surrogate key of the order
customer_id	orders	Anonymised customer identifier
order_date	orders	Date the order was placed
region	orders	Region of delivery: North, South, East, West

category orders Product category: Electronics, Home, Clothing,

		Sports, Toys
items	orders	Number of items purchased
order_value	orders	GBP value (inc. VAT & shipping)
shipment_id	shipments	Surrogate key of the parcel
courier	shipments	Arrow, Bolt, or Cyclone
ship_date	shipments	Date the parcel left the warehouse
delivered_date	shipments	Date delivered (blank → not yet delivered)
status	shipments	Delivered, Returned, or Lost
shipping_days	shipments	delivered_date – ship_date (pre-calculated)
refund_id	refunds	Surrogate key of the refund event
refund_date	refunds	Date refund processed
refund_reason	refunds	Damaged, Late, Wrong Item, Customer Cancelled
refund_amount	refunds	GBP refunded (always ≤ order_value)